Which Reroute Will an Airline Accept? Learning Route Preferences from Revisions

Ramon Dalmau[1], David Perez[2], Franck Ballerini[1], Seddik Belkoura[2],
Vincent Taverniers[2], Gratiel Marius Marin[2], Robin Deransy[1], Benjamin Cramet[3] and Grigori Gustin[3]
EUROCONTROL
[1]Innovation Hub, Brétigny-sur-Orge, France [2]Maastricht UAC, the Netherlands [3]Headquarters, Brussels, Belgium

***Abstract*—Every revision of a flight plan records a choice that an airline actually made, namely the route it dropped and the route it filed instead. Drawing on 1.48 million such pairs from eighteen months of European traffic, this paper learns which of two routes an airline prefers. Each route is described by how much more it costs than the other route of its pair and by the published points it uses; a ranking model is fitted on the first fourteen months, tuned on the fifteenth and evaluated once on the last three. It identifies the route the airline chose 68.1 % of the time, against 55.0 % for the strongest single-indicator rule, which is an inverted one preferring the costlier route. Because its confidence is calibrated, a channel proposing cheaper routes can be aimed at a precision fixed in advance, and one is simulated here against the airlines' own later filings. The paper reports what the model has learned as well as how well it scores, namely that route structure moves its decisions more than any single cost indicator does. An archive of revisions carries a bias of its own: fitted on revisions alone, the model urges change on flights that in fact kept their route, and does so more often than not. A much scarcer second set, pairing a kept route against one a similar flight flew that day, removes nearly all of it. What an airline will accept can be read from what it has already filed.**

***Index Terms*—air traffic flow management, airline preferences, learning to rank, fuel efficiency**

I. Introduction

On a spring morning in 2026, an Airbus A320 was filed from Amsterdam to Barcelona on a route carrying 102 minutes of attributed air traffic flow management (ATFM) delay, and the operator subsequently re-filed. The new route shed that delay completely, and it cost the operator on each of the three counts an airline pays for directly: 39 minutes of flight time, 1 397 kg of extra planned fuel, and 466 euros more in en-route charges. Both routes are held in the network's records and so, implicitly, is the operator's judgement that the delay would have cost it more than the detour did.

That judgement is the quantity this paper models. A flight plan is not filed once and left alone, in that an airline may revise it at any time until the aircraft leaves, and the response of the network is only one of the reasons for doing so. Attributed delay of the kind above is the most visible prompt, though not the commonest. An aircraft arriving late from its previous leg is another, since the delay it brings consumes the slack the rotation was holding, and an updated forecast is a third, since changed winds change the time and fuel every candidate route would need. The archive records what changed rather than why, and every revision leaves the same trace: the route replaced and the route that replaced it.

The scale of what is at stake is published every year. The Performance Review Commission reports how much further European traffic flies than a direct reference between the points at which it enters and leaves the airspace [1]. Not all of that distance is avoidable: airspace that cannot take all the traffic that wants it has to be protected, and a flight routed around it flies further, so part of the figure is what respecting a limited capacity costs. What this paper takes up is whether any of the rest is a cheaper route that the operator would have been willing to fly and did not file. That such routes exist is assumed here rather than shown. Candidate routes are not hard to come by. What is hard is knowing which candidate this particular operator would accept, on this day, with this rotation to protect, and knowing it while there is still time to file.

The contribution of this paper is threefold. First, it shows that airline route acceptance is learnable at network scale from revision events alone, and that the confidence of the resulting model is calibrated well enough to support a chosen operating point rather than a single accuracy figure. Secondly, it argues that the label construction is itself a design decision worth defending: a revision pair records a preference the airline actually expressed, whereas the usual alternative, which ranks the filed route above candidates produced by a route generator, rests on a comparison the operator may never have been shown. Thirdly, it reports what the model has actually learned rather than only how well it scores, namely that route structure outweighs any single cost indicator, together with the caveats that such an attribution demands.

The remainder of this manuscript is organised as follows. Section II explains why a pair of routes drawn from a revision constitutes a preference observation and what a model of such observations is for, and Section III positions the work within the literature on rerouting and route choice. Section IV describes the archive, the split protocol and the model, while Section V reports the results. Finally, Section VI concludes and outlines future work.

II. Background

This section explains how a flight-plan revision becomes a labelled preference pair, why that label is preferred to one built from generated candidates, and what the model is for.

A. Flight-plan revisions as preference data

In the European network a flight plan is filed before departure and may be revised until the flight leaves. Every filing is a message, and so is every later change, and the network keeps them all, so that what it holds for a flight is the whole sequence of messages sent for it. A route change is accordingly a pair of consecutive messages rather than a record of its own: the message that changed the horizontal route, and the one it replaced. The two together describe a decision, because the operator had committed to the earlier route and then filed the later one for the same flight on the same day, which makes the later route the one it chose and the earlier one the route it abandoned.

The sharpest of those decisions are made under flow management. When the traffic expected in a volume of airspace exceeds what can safely be handled there, whether because of demand, staffing, weather or equipment, a *regulation* is placed on it and the flights crossing it are given calculated take-off times at the rate it can absorb; the difference between a flight's calculated and requested take-off times is its attributed ATFM delay. A lightly binding regulation leaves most of its traffic undelayed, which is why a route can carry a regulation and no delay at all. An operator facing an attributed delay has a choice that is genuinely economic: accept it, or file a route that avoids the constrained volume at the cost of distance, fuel and en-route charges, and neither answer is universally correct. For instance, a flight with a tight aircraft rotation may find forty minutes of delay far more expensive than a tonne of fuel, whereas the same delay on the last leg of the day may well be cheaper than the detour. This is why the decision is worth learning rather than deriving: the trade depends on operator-specific and flight-specific costs that the network does not observe. Published European reference values [2] give the order of magnitude of the cost of delay and are widely used for analysis, but they are not the cost function any particular operator applies.

Most revisions, though, are not a response to flow management at all. Seven pairs in ten in this archive carry no regulation on either route (Table I), and within those the two routes differ by a median of one kilogramme of planned fuel. A revision made for any of the other reasons leaves the same trace as one escaping a delay.

The same events also settle how the training labels are built, because which of two routes an operator prefers cannot be asked at network scale but it can be observed, and the labels used herein follow the logic of revealed preference (i.e., inferring what an actor values from the choice it made when a concrete alternative was available) [3]. The alternative construction, adopted by the authors' own earlier work, ranks the filed plan above candidates supplied by a route generator [4]. Its weakness is not that such routes are unrealistic. It is that nobody knows whether any of them ever reached the dispatcher who planned the flight. The label asserts that the operator preferred the filed route to the generated one, whereas all the record shows is that the operator filed one route, so the comparison the label claims may never have taken place at all.

A revision pair claims nothing that did not happen, and Fig. 1 draws the two constructions side by side on one city pair. Both routes of a revision reached the dispatcher's screen, since the operator filed each of them in turn, for the same flight, on the same day. The route labelled less preferred is therefore not a plausible alternative constructed after the fact but one the operator had committed to and then abandoned, which makes the training signal a comparison that demonstrably took place.

Ranking a filed plan above generated candidates may require only that the model recognise which route looks like a filed plan, which is a different and much easier skill than knowing which route an operator would prefer, and accuracy alone cannot distinguish the two. For instance, a generated candidate may join published segments in a way no dispatcher would ever file, so that telling it from a filed plan can be done on the shape of the route alone, without knowing anything about the operator that would fly it. No dataset offers both label sources for the same flights, so the argument is about how the labels are built, not a measured comparison. The question also narrows: the model is not asked what an airline will file, but which of two specific routes it prefers, which is a strictly easier question and precisely the one a proposal channel needs answered.

B. Two operational use cases

A model of route acceptance is worth building only if some decision changes when it exists. The first such decision is a proposal channel: propose to an airline a route that is environmentally cheaper than the one it filed and that the model deems likely to be accepted, on the grounds that the operator may not have seen or evaluated that alternative in time, and it is the use case that Sections IV-B and V simulate.

The second use case is overload release: when a traffic volume is overloaded and flights may be rerouted out of it, a channel would have to decide which flights to approach first, and ranking them by whether an acceptable and cheaper alternative exists would relieve the overload while asking less of the operators concerned.

Neither use case is tested end to end here: this paper demonstrates the preference model on which both depend, and simulates the first against the airlines' later filings.

III. Related work

Work on rerouting in air traffic flow management has largely approached the problem from the network side. Over two decades ago, the authors of [5] formulated rerouting as a dynamic network flow problem, and the authors of [6] subsequently added a stochastic model that reroutes flights as capacity evolves. Both optimise an objective defined by the network and assume, in effect, that the resulting assignment is executed, so that the willingness of the airline to fly it is not modelled.

More recently, route choice in Europe was studied as a trade between fuel and en-route charges [7], which establishes

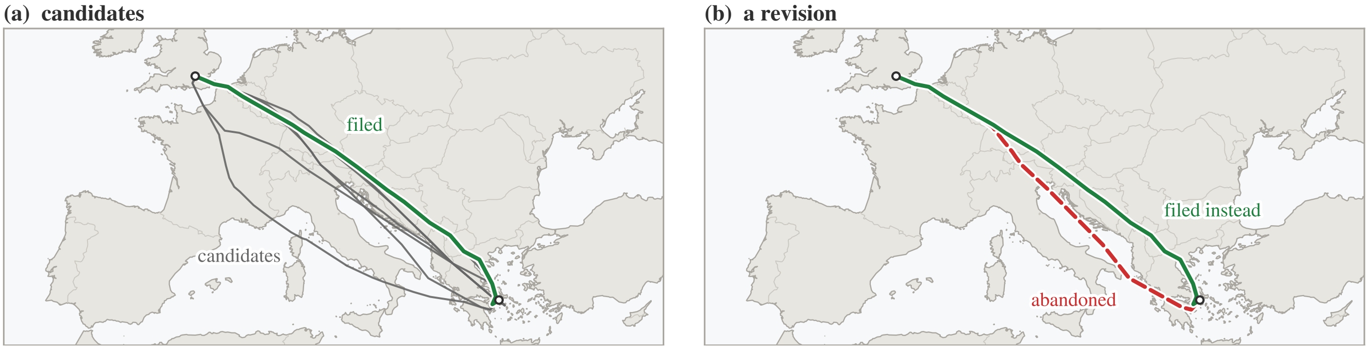

Fig. 1. Two ways to build a training pair, on one city pair. Left: the filed route (solid green) is ranked above candidates a route generator would supply (grey); the grey tracks drawn here are real routes flown on the same city pair, standing in for generated ones. Right: a revision supplies two routes that the same operator filed for the same flight hours apart, one of which it abandoned. Only the right-hand pair records a preference the airline actually expressed.

that the choice is economic and therefore learnable, but uses no label in which the operator rejected an alternative it had itself filed. The closest work to the present one is that of the authors of [8], who incorporate airline preferences into collaborative rerouting decisions and show that accounting for them changes which reroutes are selected. Those preferences, however, are inputs to an optimisation, whereas this paper learns the preference function itself, at network scale, from what airlines did rather than from what they stated.

The direct predecessor of this work applies learning to rank to flight routes for a different purpose: predicting the flight plans that have not yet been filed, so that traffic demand can be estimated earlier than the filing horizon allows [4]. That work established the framing adopted herein, namely a gradient-boosted ranker that scores routes by their indicators in context, and it reported the very difficulty that motivates the present paper. Its training pairs take the filed plan as preferred and a set of generated proposals as less preferred, which is the construction that Section II argues against.

IV. Methodology

Figure 2 shows the procedure end to end; the two subsections that follow cover the data and the model.

A. Data and pair construction

Eighteen consecutive months of European network data, from January 2025 to June 2026, are used. After retaining only those revisions that change the horizontal route, and only those pairs with complete key performance indicators, the archive contains 1.48 million labelled pairs.

What those decisions look like matters before any model touches them, and Table I breaks the archive down by the regulation context in which each revision was made. The unregulated majority differs by a median of one kilogramme of planned fuel, and the pairs that cost fuel are elsewhere: the revisions that escape a delay outright, or reduce one, together account for 18.4 % of the archive. A revision that escapes a delay moves to a route burning a median 101 kg more than the one it replaced, and that route is the cheaper of the two in only 28.1 % of cases; a revision that merely reduces the delay costs a median 65 kg and is cheaper in 31.6 %. A proposal channel is accordingly not looking for the common case, but for the minority of flights on which a genuinely cheaper alternative exists and would be accepted.

TABLE I
THE REVISION ARCHIVE BY DECISION CONTEXT, OVER ALL EIGHTEEN MONTHS. A ROUTE COUNTS AS REGULATED WHEN THE FLIGHT, WHILE FILED ON THAT ROUTE, HAD A REGULATION ATTRIBUTED TO IT, WHICH LEAVES THE DELAY ITSELF FREE TO BE ZERO. UNREGULATED PAIRS CARRY NO REGULATION ON EITHER ROUTE, ESCAPES CARRY ONE ON THE ABANDONED ROUTE ONLY, THE NEXT TWO CARRY ONE ON BOTH, AND THE LAST ONLY ON THE NEWLY FILED ROUTE. THE FUEL DIFFERENCE IS BETWEEN THE TWO ROUTES OF A PAIR.

context	pairs	share [%]	median Δfuel (new − old) [kg]	new route burns less [%]
unregulated	1 047 228	70.6	1	46.2
escaped delay	117 594	7.9	101	28.1
reduced delay	155 326	10.5	65	31.6
kept or raised delay	141 648	9.6	0	47.4
took on delay	21 337	1.4	−5	51.8

Each route is described by twenty-three features, summarised in Table II. The two routes of a pair are separated in time as well as in geography, however, and that must be addressed before anything is learned. The replacement is by construction the later message, so any feature that drifts with time identifies the answer without saying anything about the quality of the route. The time remaining to departure is the clearest case: it is necessarily smaller for the route that replaced the other, so a model given it in raw form could answer “the one filed later”, be right on essentially every pair, and know nothing whatever about routes. This is leakage rather than learning, and it would be invisible in the accuracy figure. To ensure a fair comparison between the two routes, every flight-level attribute was therefore harmonised across the pair, both routes taking the value carried by the later message, so that no such attribute can separate them and only route-level quantities differ. The filing time is the case that matters most, and harmonising it in that way leaves both routes carrying the smaller of the two times to departure, which is what makes the shortcut disappear.

Two feature-construction decisions shaped the result more than the choice of learner did, and the first is that the four computed indicators enter as within-pair differences rather than magnitudes, obtained by subtracting the pair minimum from each, because absolute quantities are close to meaningless across the network: a 2 000 kg fuel figure means one thing on a short-haul leg and quite another on a long-haul sector, whereas the 300 kg by which one route exceeds the other carries the same meaning whether the sector is 400 or 4 000 km. Secondly, the ordered waypoint sequence is carried as text and encoded as a bag of tokens (i.e., each waypoint identifier is treated as a word and the route as the unordered collection of its words), which lets the model exploit route structure rather than aggregate distance alone. For instance, two routes on the same city pair may differ by under one per cent in distance and in planned fuel, and so be indistinguishable on every indicator, while only one of them crosses a volume whose identifier the model has met in thousands of earlier revisions; the token set carries that fact and the aggregate distance cannot. Section V-B shows that the first of the two matters far more than the authors expected.

Every revision pair is, by construction, a flight that *did* change route, and that creates a trap. A model trained only on such events learns that change is what happens, and recommends it far too eagerly against traffic that mostly stays, a bias that Section V-B measures. The archive can say which route a flight moved to, and nothing at all about the route a flight declined by staying where it was, because staying leaves no record of the alternative, so that record has to come from somewhere else.

A second, much smaller dataset was therefore constructed, in which *keeping* the route is the revealed choice. Let us consider two flights on the same day, with the same origin, destination and aircraft type, that flew different routes and neither of which revised its plan, and let us suppose that one of them was caught by a regulation while the other was not. The regulated flight had a reason to move and an alternative that was demonstrably flyable that day, namely the route its twin was flying, and yet it stayed, though the twin does not establish that the same routing was open at the stayed flight's own departure hour. Its route is accordingly labelled as preferred, and that of its twin as the alternative it declined. The construction carries a confound that has to be stated with it: the regulated flight is always the one whose route was kept, so regulation status identifies the labelled answer. The conditions are restrictive and the yield is correspondingly low: 6 672 such *stay pairs*, of which 1 093 fall in the test months, against 1.48 million revision pairs. The two are used together, with stay pairs entering the fitting set at an increased group weight (the pair's weight in the ranking loss).

Whether the model takes the shortcut that confound offers can be tested, because the archive holds the mirror population. In an *escape* the regulated route is the one the flight *abandoned*, so a rule keyed on regulation status alone, which is right on every stay pair by construction and without examining a route at all, is wrong on every escape. Section V-B puts the two populations side by side.

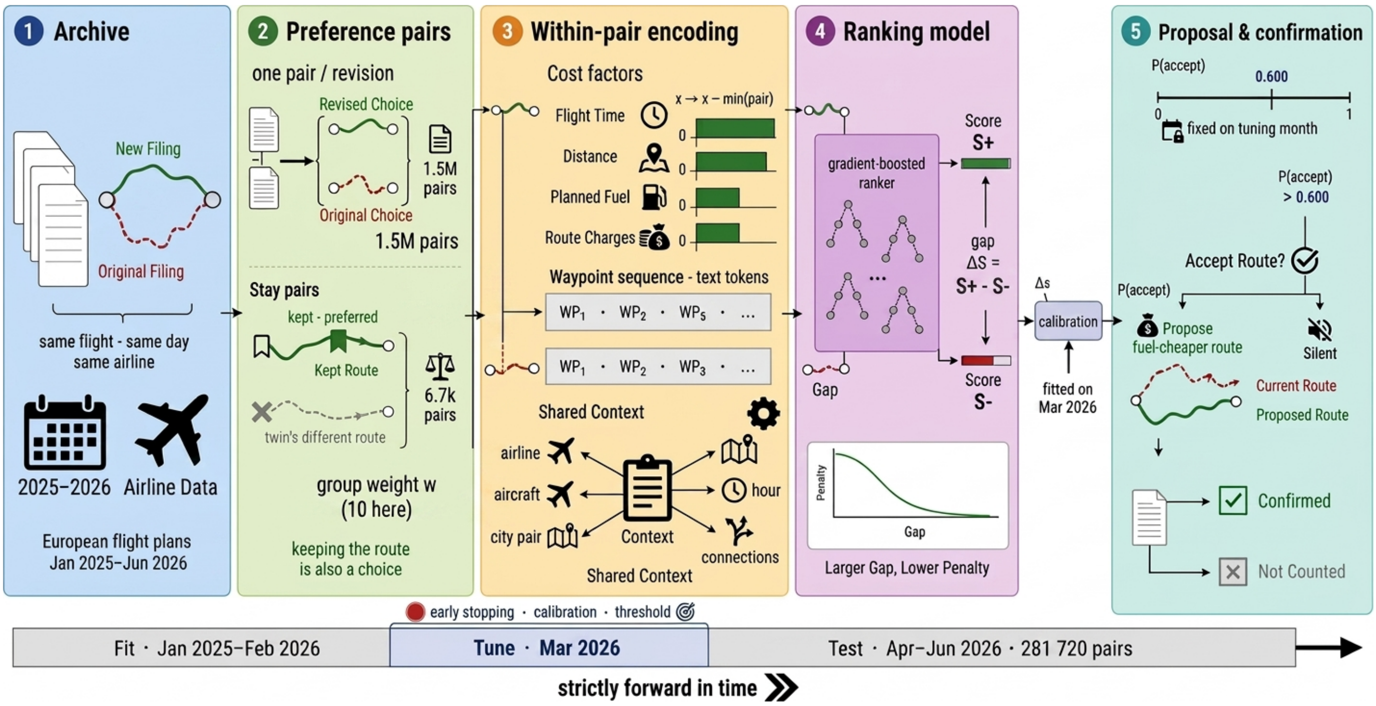

Fig. 2. How a filing archive becomes a proposal, above the fit, tune and test periods. The pair box shows both constructions: a revision, which is one flight and two routes, and a stay pair, which is two flights. The inset in the fourth stage is the training objective, whose penalty falls as the gap between the two scores widens and never reaches zero; the calibrated probability against its threshold is the only condition on a proposal. The bars of the third stage are one illustrative pair, in which the chosen route is the costlier on each indicator, the commoner direction on this archive rather than the invariable one.

TABLE II

WHAT THE MODEL SEES FOR EACH ROUTE; SOME ROWS SUMMARISE SEVERAL FEATURES (TWENTY-THREE IN ALL). THE FIRST TWO GROUPS AND THE LAST BELONG TO THE ROUTE AND DIFFER WITHIN A PAIR, THE INDICATORS ENTERING AS DIFFERENCES FROM THE PAIR MINIMUM; THE ROTATION AND CONTEXT ROWS BELONG TO THE FLIGHT AND CARRY THE SAME VALUE ON BOTH ROUTES, THE ONE EXCEPTION BEING THE OUTBOUND CONNECTION TIME, WHICH IS SHORTENED ON THE SLOWER ROUTE BY THE DIFFERENCE IN FLIGHT TIME AND IS THEREFORE ROUTE-DEPENDENT AFTER ALL; THAT ADJUSTMENT FOLLOWS FROM THE ROUTE'S OWN FLIGHT TIME AND CARRIES NOTHING ABOUT WHICH MESSAGE WAS FILED LATER.

group	content
indicators	flight time, distance, planned fuel, route charges
regulation	attributed delay (own, inbound, outbound), their count, reason, and the type and identifier of the protected volume, meaning the airspace whose capacity the regulation protects
rotation	inbound and outbound connection times and states
context	operator, aircraft type, city pair, hour, weekday, month, time to departure
route	the ordered waypoint sequence, as text

The evaluation is strictly out of time, in the sense that the model is fitted on January 2025 through February 2026 (1 134 064 pairs), tuned on March 2026 (67 349), and evaluated once on April to June 2026 (281 720). Every threshold, calibrator and stopping decision is taken on the tuning month alone, and flights appearing in the tuning or evaluation months are removed from the fitting set. All the data used is held by the network for operational purposes and was processed under existing data-handling arrangements. Operator identity enters the model as a categorical feature only, and no operator or airspace volume is named anywhere in this paper.

B. Model, training and calibration

The operational question is comparative: given two routes for one flight, which of them does the airline prefer? Accordingly, the model gives each route a single number, which is not a probability and has no units, and which is only meaningful next to the number given to another route for the same flight. Training pushes those numbers apart in the right direction. For every historical pair the model pays a penalty, largest when the abandoned route outscores the chosen one; the penalty falls as the chosen route's lead grows and never quite reaches zero, so the model keeps pressing even once the order is right [9]. Nothing in the objective asks the model to reproduce a route in isolation, which is what makes the pair the natural unit.

Gradient-boosted decision trees are used as the scoring function [10], at a depth of 8, with the learning rate selected automatically and with early stopping on the tuning month. Stay pairs enter the fitting set with a group weight w, swept over $\{10, 30, 100\}$; Section V-B reports why 10 is used.

Once trained, the model has still to be connected to a decision. Let us consider a real pair from the held-out quarter, on which the model scores the two routes -2.82 and $+0.90$. Neither number means anything on its own, since the score has no units, and the same two numbers on another flight would carry no comparable meaning. What can be read is their difference, here 3.7 in favour of the second route. Even that is not yet a decision, because nobody in an operations room knows what a gap of 3.7 is worth.

Calibration is the step that turns the gap into a statement of the form "about eight times in ten, the route the model prefers is the one the airline goes on to file". It is a monotone mapping from score gap to the observed rate at which the preferred route was the one filed, fitted on the tuning month alone [11] and applied unchanged to the evaluation months. Isotonic regression was selected there on log loss, against the raw sigmoid of the score gap and a two-parameter Platt fit, before those months were touched.

Models of this kind usually over- or under-state their confidence and benefit from such a fit [12]. This one did not. On the held-out quarter its own raw sigmoid was better calibrated than the isotonic fit chosen to correct it: the sigmoid missed the observed rate by 0.9 percentage points on average, the isotonic fit by 1.8. The choice was fixed before those months were opened and was not revised afterwards, since revising it on held-out evidence is exactly the peeking the protocol forbids, so on this archive the fitted mapping is a safeguard rather than a correction. What matters operationally is that stated and observed rates agree to within those 1.8 points, because without calibration one can rank proposals but cannot decide how many of them to make.

The policy that results is short. A flight is eligible whenever its two routes differ in planned fuel at all, which holds for 97.9 % of the held-out pairs, and the cheaper of the two is the candidate. That candidate is proposed when its calibrated acceptance probability exceeds a threshold τ, selected on the tuning month for a target precision and reported unchanged on the evaluation months; $\tau = 0.600$ throughout what follows. The calibrator always scores the route the model prefers, so the probability it returns is by construction at least a half. Where the cheaper route is the one the model ranks lower, the channel must therefore use one minus that probability, which is below a half, so such a pair is never proposed at any threshold of a half or above. A proposal is realised when the airline's own subsequent filing chose that route.

Note that the cheaper route is not usually the one the airline goes on to file. In 55.1 % of eligible pairs it is the route being abandoned, so a channel proposing the fuel saver on every eligible flight would be right 44.9 % of the time on those eligible pairs, which is worse than a coin toss and is the figure Section V measures the channel against. The same rule scores 45.0 % across all held-out pairs, a difference of population and not of rule.

V. RESULTS

The model is reported first on how often it is right, then on which choices in its construction earned that, then on what it has learned and where it fails, and last on what a channel built upon it would deliver.

A. Predictive performance

On the held-out quarter the model identifies the airline's chosen route in 68.1 % of 281 720 pairs, against the 50 % of

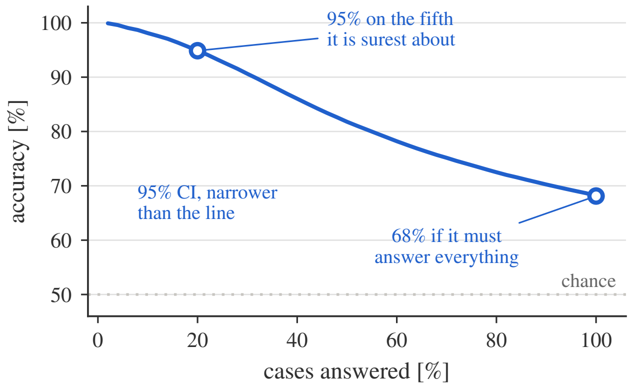

Fig. 3. Accuracy against the share of cases answered, on the held-out quarter, a case being one pair of routes. Moving left, the model answers fewer cases, lowest confidence dropped first, and is right more often; the dotted line is chance.

random choice. The 95 % interval is 67.9 to 68.3, widening to 67.7–68.6 when flights on the same city pair are counted as one correlated group, and March, the tuning month, scores 67.5 %, so there is no sign of overfitting to it.

Simpler methods on the same quarter are the more informative reference, and every rule preferring the cheaper route scores below chance: the fuel rule is right 45.0 % of the time, so the strongest single-feature rule is its inverse, "prefer the route that burns more planned fuel", at 55.0 %. The most accurate rule of the kind is a different one, preferring the route that carries less attributed delay, and on the pairs it can decide it is right 71.6 % of the time. Those are few. An unregulated route has no delay figure at all rather than one of zero, and the rule can separate two routes only when their delay figures differ, which together leave it 14.6 % of the pairs; scoring the remainder as coin tosses leaves the delay rule at 53.2 % across the archive, below the inverted fuel rule.

On the fifth of the pairs it is most confident about, the model is right 94.9 % of the time. Fig. 3 traces the whole of that trade, in that the model is permitted to abstain on the cases in which the two routes score close together, so that the share of cases it answers falls as its accuracy on them rises. A channel need not answer every flight, and accuracy alone therefore understates what the model is worth.

The model therefore stands 13.1 points above the best single-indicator rule, 68.1 % against 55.0 %, which establishes that route choice carries structure that no single indicator recovers.

Where those 68.1 % sit is more informative than the pooled figure, and Table III splits the quarter three ways. Accuracy climbs steeply with the delay carried by the abandoned route, from 63.9 % where there was none to 84.5 % between half an hour and an hour, where the climb stops. That is what an economic reading predicts, since the larger the delay, the clearer the incentive and the easier the decision is to predict.

The third block of the table repays more attention, because it inverts the expectation. The model is right 79.9 % of the time on the pairs where none of the three costs an airline pays for directly, namely flight time, planned fuel and route charges, is cheaper on the newly filed route, and only 53.9 % on those where all three are. That first row is not a population that moved to a strictly worse route, since a pair whose two routes are identical on an indicator falls into it as well, and on the 42.9 % of it that is strictly worse on all three the model is right 86.3 % of the time.

TABLE III

Accuracy by stratum on the held-out quarter, with Wilson 95 % intervals. The three blocks split the same 281 720 pairs three ways, the last of them by how many of the three costs an airline pays for directly, namely flight time, planned fuel and route charges, are strictly cheaper on the newly filed route, so that an indicator on which the two routes are identical, as route charges are on 46.4 % of the quarter, counts towards the first row rather than the last.

stratum	pairs	accuracy [%]
which route carried a regulation		
neither	178 370	64.0 [63.8, 64.2]
old route only	30 953	88.0 [87.6, 88.3]
both	67 587	70.6 [70.2, 70.9]
new route only	4 810	58.6 [57.2, 60.0]
delay on the abandoned route		
none	205 205	63.9 [63.7, 64.1]
up to 15 min	19 672	70.3 [69.7, 71.0]
15 to 30 min	18 283	79.3 [78.7, 79.9]
30 to 60 min	21 231	84.5 [84.0, 85.0]
above 60 min	17 329	83.5 [82.9, 84.0]
cost indicators favouring the new route		
0 of 3	90 961	79.9 [79.7, 80.2]
1 of 3	87 471	70.0 [69.7, 70.3]
2 of 3	78 552	56.8 [56.5, 57.2]
3 of 3	24 736	53.9 [53.3, 54.5]

Those two rows are 26.0 points apart, and the archive explains the gap only in part. A move the indicators all argue against is more often a move made for a reason they do not carry, and a regulation is attributed to one of the two routes on 42.2 % of the first row against 33.7 % of the last. Those 8.5 points of difference in regulation exposure are not enough to account for 26.0 points of accuracy, and nothing else in the archive connects the two. The direction is nevertheless not in doubt: the model contributes most where cost logic fails, which is the right division of labour for a system meant to complement cost models rather than replace them.

A model that scores well on average but decays from month to month is not deployable, and across April, May and June accuracy moves between 67.5 % and 69.0 %. A quarter of evidence rules out rapid decay, but it does not establish the retraining cadence a deployed system would need, which the journal extension measures directly.

B. Ablation and change bias

Table IV reports screening ablations, each configuration changed in turn and the model retrained, and one of the five dominates the rest.

Expressing the indicators relative to the other route is that one: removing it costs nearly ten percentage points, far more than any other component, because it is what converts four absolute magnitudes into a comparison.

TABLE IV

SCREENING ABLATIONS. EACH VARIANT IS RETRAINED FROM SCRATCH AT A REDUCED SETTING (DEPTH 6, 800 ROUNDS) THAT REACHES 60.6 % ON REVISION PAIRS; VALUES ARE THE CHANGE IN REVISION ACCURACY AGAINST THAT BASELINE, NOT HEADLINE ACCURACIES. A POSITIVE VALUE MEANS THE VARIANT SCORES *higher* ON REVISIONS, WHICH FOR THE STAY-PAIR CORRECTION IS THE PRICE PAID FOR THE CHANGE-BIAS REDUCTION THE TEXT REPORTS NEXT. EVERY DIFFERENCE IS SIGNIFICANT UNDER MCNEMAR'S TEST AT $p < 0.001$ EXCEPT THE WAYPOINT TEXT ($p = 0.58$). THE MONOTONE CONSTRAINT FORCES THE SCORE NEVER TO RISE WITH THE NUMBER OF REGULATIONS ON A ROUTE; A VERTICAL-ONLY REVISION LEAVES THE HORIZONTAL ROUTE UNTOUCHED, SO ITS TWO ROUTES ARE IDENTICAL AND THE PAIR UNANSWERABLE.

variant	Δ accuracy [pp]
indicators absolute, not within-pair	−10.0
without stay-pair correction	+3.7
with monotone constraint	−0.2
without waypoint text	+0.03
including vertical-only revisions	−2.0

The waypoint text is worth 1.4 percentage points at full scale, 68.2 against 66.8, the 68.2 being a separate run and the seeds of the adopted model spanning 0.17 points. Interestingly, at screening scale the same feature appears worthless, the reason being that a bag-of-words representation spread over thousands of rare tokens cannot repay its cost in 800 shallow rounds. The row therefore illustrates why screening ablations can mislead, and the feature is a real but secondary contributor, worth about a tenth of the model's margin over the best single rule.

On the 1 093 stay pairs of the held-out quarter, a model trained on revisions alone is right 34.8 % of the time; adding stay pairs at group weight 10 raises that to 90.0 %, the largest correction any of these choices makes. The 34.8 % is the change bias of Section IV measured: revisions teach the model that a regulated route is one to escape, so it carries that lesson into a population where the opposite holds and prefers the alternative route in 65.2 % of the cases where the flight demonstrably kept its own. The correction costs 0.8 points on the revision metric at full scale, 68.1 % against 68.9 %, the 3.7 points of Table IV being measured at the reduced screening setting.

The same model scores 90.0 % on the 1 093 stay pairs of the held-out quarter and 88.0 % on its 30 953 escapes, and no rule keyed on regulation status alone can produce that pair of figures, since such a rule is right on every stay pair and wrong on every escape by construction. Telling the two situations apart takes the route and the flight around it, so the 90.0 % is not the model reading the confound back off the construction.

The weight $w = 10$ is used here because it takes nearly all of the available correction at the smallest cost, and because the tuning month selects it as well.

C. Error analysis and feature attribution

Fortunately, errors concentrate where the model is already unsure. Only 5.6 % of them fall among the quarter of pairs whose two routes score furthest apart, against the 25 % that would fall there if error were independent of confidence, and

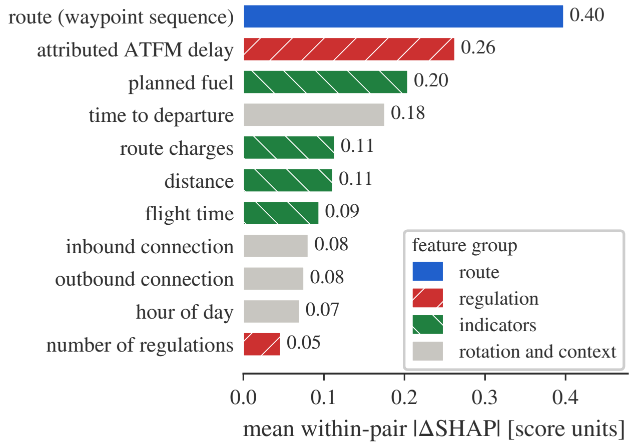

Fig. 4. What moves a decision: the mean absolute *difference* between a feature's Shapley value on the two routes of a pair, over the held-out quarter, with the rotation and context groups of Table II drawn together. The waypoint sequence is the largest single term, ahead of every *individual* cost indicator, though the four taken together sum to more; attributed ATFM delay outweighs planned fuel. Rotation and context features are identical on both routes, so their bars measure interaction rather than preference.

27.7 % are near-ties in which the two scores land within one fourth of the archive's median gap. The expensive errors are the opposite ones, in which the model confidently proposes a route that the operator declines, and the change-bias measurement above is the closest available proxy for their rate.

A ranking model an operational user is asked to trust must be open to interrogation, and Shapley attributions were therefore computed over the held-out quarter [13]. Because the model decides on a comparison rather than on one route, what Fig. 4 reports is the *difference* between a feature's attribution on the two routes of a pair, averaged in magnitude over the quarter. That is a ranking not of what correlates with acceptance, but of what moves this particular model's decisions.

On that scale the waypoint sequence moves the score by 0.40, attributed ATFM delay by 0.26 and planned fuel by 0.20, so route structure carries information about the choice that the cost indicators do not.

The waypoint text ranks first by attribution, yet removing it and retraining costs only 1.4 points of accuracy, because attribution measures how much a feature moves the score of the model at hand while ablation measures how much worse a model without it would be.

D. What a channel would deliver, and the conditions for it

One proposal says little about a channel making tens of thousands, and three conditions stand between these results and an operational channel. Attributed delay is among the strongest inputs and is a property of a moment rather than of a flight, so the first constraint on a deployed channel is the freshness of the regulation picture rather than the cost of computation. The population changes next, in that a channel would see every flight and most never revise their plan at all.

Acceptance itself, finally, is never observed here at all, every figure resting on what the airline filed next.

At an operating point fixed in advance for 80 % precision ($\tau = 0.600$, delivering 79.2 %), the simulated channel surfaces 56 162 proposals over the held-out quarter carrying 20.1 kt of planned fuel, counted only where the airline's own later filing chose the route proposed to it, or 63.5 kt of CO_2 at 3.16 kg per kilogramme of fuel, and between 222 and 254 kt of CO_2 a year: the lower end applies the per-eligible-flight rate to a year's eligible traffic, and the upper multiplies the quarter by four. These figures are computed on flights that revised their plans, and planned fuel is not burned fuel, so they should be read as what a channel could have surfaced earlier rather than as a traffic-wide counterfactual. The full channel analysis is the subject of a journal extension of this work.

VI. Conclusions

Airline route acceptance is learnable from flight-plan revisions at network scale, and the findings of this paper are threefold. First, a model fitted on 1 134 000 revisions identifies the chosen route in 68.1 % of held-out cases, 13.1 points above the best rule built on any one indicator, and its confidence is informative enough to choose where to operate rather than settle for a single accuracy figure: on the fifth of decisions it is most confident about, it is right 94.9 % of the time. Secondly, that margin is widest exactly where cost logic fails, 79.9 % where no cost indicator favours the route the airline moved to against 53.9 % where all three do. Thirdly, the model weighs route structure more heavily than any individual cost indicator, though the four cost indicators together outweigh it and removing the waypoint text costs only 1.4 points, so what it contributes is real but secondary. It is nonetheless information that a formulation based on costs alone cannot represent.

The number that characterises this system is not 68.1 % but the curve of Fig. 3, because a channel chooses where on that curve to sit, and because ranking a pair correctly and proposing correctly are different quantities of which the second is the operational one: at the threshold used here, the channel proposes on about a fifth of the flights whose two routes differ in planned fuel at all, and 79.2 % of those proposals match what the airline went on to file. Accordingly, a first deployment would sit at low coverage on Fig. 3, answering only the cases it is most confident about, and would widen that coverage only as acceptance is observed.

Two bounds remain. The first is the population the second deployment condition names: the data on flights that stayed is small, 6 672 pairs in all and 1 093 in the test months, and pairs two different flights rather than two routes for one, so a model could learn to recognise the construction rather than the preference it encodes. The 90.0 % against 88.0 % of Section V-B rules out a model reading regulation status and nothing else, but it cannot rule out every cue the construction leaves behind.

The second lies in how a route is represented. The model reads a route as the published identifiers along it, so what it holds about route structure is a vocabulary rather than a geometry. Those identifiers are European, so little of the 1.4 points the waypoint text is worth would survive a move to another region, and they are revised from time to time, so a piece of airspace can return under a name the model has never met. The obvious remedy was tried and did not work. Recast as a sequence of descriptors that depend on its shape alone and passed through a neural encoder, a route scored worse than it did by its tokens, and a linear classifier on the encoder's own representation separated preferred from abandoned routes at 0.54 against the 0.50 of chance, which points at the geometric input rather than the objective as the limitation. A name-independent representation of route shape is nonetheless what would make this model portable, and remains worth pursuing.

Future work should therefore be directed, in the first place, at assembling a balanced stay dataset in which regulation status is independent of the label, which requires data that is not currently assembled rather than data that does not exist. Last but not least, a shadow-mode trial logging the real responses of operators would close the remaining gap.

Disclaimer

The views expressed are those of the authors and do not necessarily reflect the official position of EUROCONTROL or of its Member States. This paper reports research and does not commit to any operational service.

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